

Volume 22

Sentinel Evidence & Forensics Framework (SEFF)

Version: 1.0

Status: Draft

Vision

Every observation inside Project Sentinel becomes evidence.

Evidence is:

- Verifiable
- Traceable
- Immutable (where appropriate)
- Searchable
- Versioned
- Explainable
- Linked to objects
- Linked to missions
- Linked to findings

Evidence is never separated from the object that produced it.

Chapter 1

Evidence Lifecycle

Observation

Everything starts with evidence.

Chapter 2

Evidence Types

Project Sentinel shall support structured evidence types including:

Network Evidence

- DNS response
 - HTTP response
 - TLS certificate
 - HTTP headers
 - Redirect chains
 - Open ports
 - Protocol negotiation
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Endpoint Evidence

- Process metadata
 - Running services
 - Scheduled tasks
 - Startup entries
 - Installed software inventory
 - Security configuration
 - File metadata
 - File hashes
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Mobile Evidence

- Installed applications
- Application permissions
- Device encryption status
- Operating system version
- Security patch level
- Configuration status

Within platform-supported APIs and user-granted permissions.

Cloud Evidence

- IAM configuration
 - Storage configuration
 - Security groups
 - Firewall rules
 - Public exposure
 - Identity policies
-

Identity Evidence

- MFA status
- Password policy
- Group memberships

- Privilege assignments
 - Login history (where available)
 - Service account configuration
-

Chapter 3

Evidence Object

Every evidence item becomes its own object.

Example

Evidence ID

Evidence is first-class.

Chapter 4

Evidence Integrity

Every evidence object stores:

Collection Time

Collector

Collection Method

Hash

Version

Digital Signature (optional)

Verification Status

Integrity must always be demonstrable.

Chapter 5

Chain of Custody

Where organizations require it, Project Sentinel records:

Who collected evidence.

Who viewed it.

Who exported it.

Who approved its use.

Who modified associated metadata.

Evidence itself is preserved.

Chapter 6

Evidence Relationships

Evidence never exists in isolation.

Example

Website

The Knowledge Graph links all supporting observations.

Chapter 7

Evidence Confidence

Each evidence source is evaluated independently.

Example:

- Direct API response → High confidence.
- Verified system API → High confidence.
- Passive observation → Medium confidence.
- Incomplete observation → Lower confidence.

The confidence reflects the quality of the evidence, not the severity of the issue.

Chapter 8

Evidence Timeline

Evidence can be replayed.

Example

January

Users can observe security improvements over time.

Chapter 9

Evidence Explorer

A dedicated interface.

Search by:

- Object
- Mission
- Date
- Evidence Type
- Confidence
- Collector
- Tags
- Hash
- User

Everything remains searchable.

Chapter 10

Evidence Export

Supported export formats:

- PDF
- JSON
- CSV
- HTML

Optionally include hashes, timestamps, and metadata so recipients can verify integrity.

PS-ADR-0019

Evidence Is the Source of Truth

Status: Accepted

Decision

Every security conclusion inside Project Sentinel shall be traceable to one or more evidence objects.

Rationale

Evidence-first design improves transparency, auditability, and user trust.

NEW CORE ENGINE

Evidence Correlation Engine 2.0

Responsibilities:

- Merge duplicate observations.
- Detect conflicting evidence.
- Rank evidence quality.
- Link evidence to related objects.
- Maintain evidence history.

This improves reasoning while preserving original observations.

NEW CORE ENGINE

Evidence Replay Engine

A unique capability.

Users can replay the exact evidence collected during a mission to understand how conclusions were reached and to compare historical states.

NEW CORE ENGINE

Evidence Integrity Monitor

Continuously verifies:

- Hash consistency.
- Metadata integrity.
- Missing evidence.
- Broken relationships.
- Storage health.

Alerts are generated if integrity checks fail.

NEW IDEA

Cyber Security Flight Recorder

Inspired by aviation.

Project Sentinel continuously records operational events during a mission.

If a major incident occurs, investigators can replay:

- Discoveries.
- Scan execution.
- Evidence collection.
- Correlation.
- Decisions.
- User actions.
- Remediation.
- Verification.

The Flight Recorder becomes invaluable for incident response and post-incident reviews.

NEW IDEA

Sentinel Evidence Notebook

Every mission automatically creates a structured notebook where analysts can:

- Add observations.
- Attach evidence.
- Record hypotheses.
- Document conclusions.
- Link related objects.
- Track investigation progress.

This combines technical evidence with human expertise, making investigations easier to understand and audit.

NEW IDEA

Evidence Trust Index (ETI)

Every mission receives an **Evidence Trust Index** based on factors such as:

- Coverage of the assessed environment.
- Diversity of evidence sources.
- Freshness of collected data.
- Integrity verification.
- Correlation quality.
- Verification success.

The ETI indicates how well-supported the mission's conclusions are, helping users understand the reliability of the assessment.

Looking Beyond Volume 22

The next engineering milestone is the **Web Security Engine**, one of the flagship components of Project Sentinel.

Unlike many scanners, it will not simply enumerate issues. It will be organized around reusable analysis engines such as:

- HTTP Analysis Engine.
- TLS Analysis Engine.
- Header Analysis Engine.
- Content Security Policy Analyzer.
- Authentication Assessment Engine.
- Technology Fingerprinting Engine.
- API Assessment Engine.
- Passive Discovery Engine.
- Authorized Active Assessment Engine.
- Screenshot and Visual Change Engine.
- JavaScript Analysis Engine.
- Evidence Collection Layer.
- Trust Evaluation Layer.

Each engine will produce structured evidence, update the Knowledge Graph, and contribute to the overall Trust Profile.

A final proposal for our engineering philosophy

I believe we should adopt one overarching design rule that governs every implementation decision:

Project Sentinel never hides complexity—it organizes it.

Internally, the platform can be extremely sophisticated, but externally it should present users with clear missions, evidence-backed findings, understandable recommendations, and guided workflows.

If we remain faithful to that principle, we can build a Cyber Security Operating Environment that is powerful enough for security professionals while remaining approachable for home users, consultants, enterprises, financial institutions, and government organizations.

====[USER]====

Good and continue

====[ASSISTANT]====

Excellent. I think we are now starting the phase that will ultimately define Project Sentinel's reputation.

People will not recommend Project Sentinel because of its UI.

They will recommend it because it **finds what others miss**, explains why it matters, and helps users fix it.

That requires an engineering-first approach.

One important point before we continue:

When we implement the Web Security Engine, we will ensure it supports **only authorized security testing**. Features that actively probe or test targets will be designed for environments where the user owns the systems or has explicit permission, including approved bug bounty programs that permit those techniques. This keeps the

platform professional, legally compliant, and trustworthy.

PROJECT SENTINEL BLUEPRINT

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